

Multilift MD1, MDV

Installation and operating instructions



Installation and operating instructions

<http://net.grundfos.com/qr/i/96102304>

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1. General information



Read this document before you install the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1.1 Symbols used in this document



DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.



WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.



CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The text accompanying the three hazard symbols DANGER, WARNING and CAUTION is structured in the following way:



SIGNAL WORD

Description of the hazard

Consequence of ignoring the warning

Action to avoid the hazard.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.



Observe these instructions for explosion-proof products.

2. Receiving the product

2.1 Transporting the product



Do not stack the product.



WARNING

Falling objects

Death or serious personal injury

- Secure the product during transportation to prevent it from tilting or falling down.



CAUTION

Back injury

Minor or moderate personal injury

- Use lifting equipment.



CAUTION

Crushing of feet

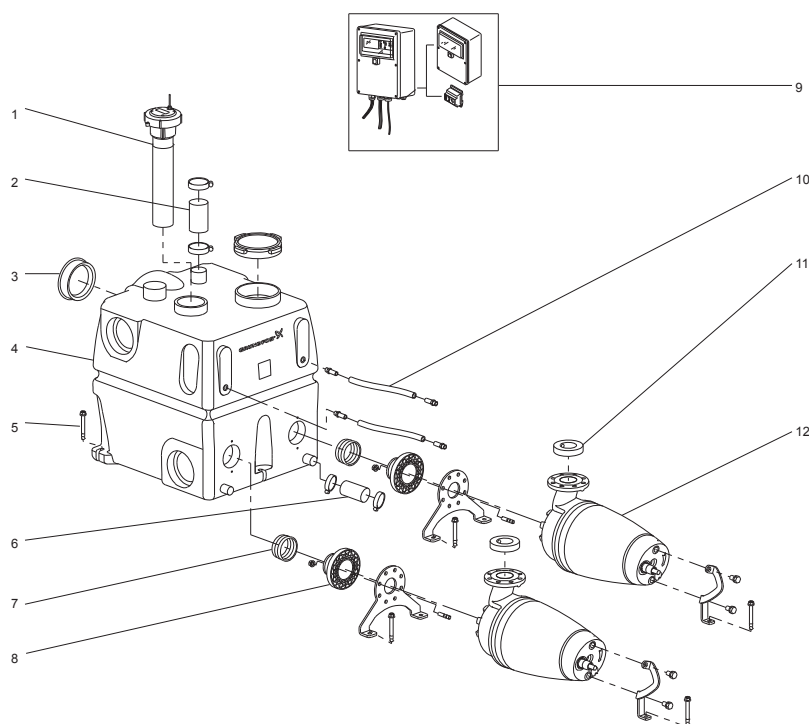
Minor or moderate personal injury

- Wear safety shoes when moving the product.

2.2 Scope of delivery

This list applies to a standard lifting station with one tank. Additional components are included if you have ordered an additional tank.

Pos.	Quantity	Description	Specification
1	1	Pressure tube	With 10 m hose
2	1	Flexible hose with two clamps for the vent pipe	DN 70
3	1	Inlet seal	DN 150
4	1	Tank	
5	3	Screws and expansion anchor for tank fixation	
6	1	Flexible connecting piece for optional diaphragm pump and fittings	For 1 1/2" pipe
7	2	Seal for pump inlet	DN 100
8	2	Connecting flange with stub	DN 80 or DN 100
9	1	LC 221 controller	
10, 11	2	Vent flange with vent hose and fitting	DN 80 or DN 100
12	2	SE or SL pump	With 10 m cable
	2	Gasket kit	DN 80 or DN 100, 8 bolts M16, nuts, washers and gaskets
	1	Installation and operating instructions	For lifting station and controller
	1	Safety instructions	For lifting station and controller



Related information

- [5.5.1 Accessories](#)

3. Installing the product

3.1 Installing the controller

3.1.1 Location

Install the product in a location that meets the following requirements:

- Place the product in a flood-safe place.
- Make sure that the ambient temperature is within the limits.
- Install the product as close as possible to the lifting station.
- If you install the product outdoors, place it in a protective shed or enclosure, class IP55.
- Do not expose the product to direct sunlight.

Related information

- [13.1 Operating conditions](#)

3.1.2 Mechanical installation



CAUTION Back injury

Minor or moderate personal injury

- Use lifting equipment.



CAUTION Crushing of feet

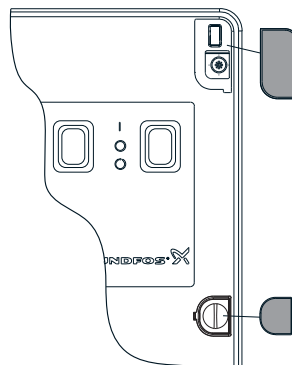
Minor or moderate personal injury

- Wear safety shoes when moving the product.

3.1.2.1 Installing the controller

You can install the controller without removing the front cover.

1. Mark up the holes on a plane wall.
2. Drill the holes.
3. Remove the plastic caps that cover the screw holes in the controller or open the control cabinet.
4. Install the controller with the cable entries pointing downwards.
If required, fit additional cable entries in the bottom plate of the control cabinet.
5. Fit the screws.
6. Fit the plastic caps, if any.



3.1.3 Electrical connection



DANGER Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product. Make sure that the power supply cannot be accidentally switched on.

3.1.3.1 Connecting the mains supply to systems with direct-on-line starting

- Make sure that the supply voltage and frequency correspond to the values stated on the nameplate on the lifting station.
- If required according to local regulations, install an external all-pole main switch.
- If required according to local regulations, install a backup fuse. See the controller nameplate.

Systems with direct-on-line starting have a 1.5 m power supply cable connected in the control cabinet. The cable is equipped with a CEE plug.

1. Install a CEE power socket less than 1.5 m from the control cabinet.
2. Connect the CEE plug to the CEE power socket.

Related information

- [6.2.1 Nameplate](#)

3.1.3.2 Connecting the mains supply to systems with star-delta starting

- Make sure that the supply voltage and frequency correspond to the values stated on the nameplate on the lifting station.
- If required according to local regulations, install an external all-pole main switch.
- If required according to local regulations, install a backup fuse. See the controller nameplate.

Systems with star-delta starting have an all-pole main switch on the cabinet front.

1. Set the main switch to off.
2. Connect a power supply cable to the control cabinet through the cable entries and gaskets. See the wiring diagram.
3. Set the motor-protective circuit breakers in the control cabinet.

Related information

- [6.2.1 Nameplate](#)

3.2 Installing the lifting station

3.2.1 Location

Install the product in a location that meets the following requirements:

- Make sure that the location is properly lit.
- Make sure that the location is properly vented.
- Make sure that there is at least 60 cm free space around the product.
- We recommend that you install a drainage pump in a pump sump below floor level if you install the lifting station in a basement with the risk of penetrating groundwater. In some countries, a drainage pump or only a pump sump is required. We also recommend that you install a separate float switch next to the lifting station and connect it to terminals 13 and 14 of LC 221. If the float switch is activated, the controller gives an acoustic alarm, the display shows the external alarm **EXTERN**, a red LED flashes, and the common fault relay is activated.



The lifting station may be flooded maximum 2 metres for seven days.

3.2.2 Mechanical installation



CAUTION

Back injury

Minor or moderate personal injury

- Use lifting equipment.



CAUTION

Crushing of feet

Minor or moderate personal injury

- Wear safety shoes when moving the product.

3.2.2.1 Installing the pumps

The description below is for a system with isolating valves and a non-return valve.



Make sure that the weight from the inlet, outlet and vent pipes does not rest on the tank. Long pipe sections and valves must be supported.

Torques for flanges +/- 5 Nm

Nominal diameter	Screws	Torque [Nm] ^{*1}	Torque [Nm] ^{*2}
DN 80	8 x M16	70	60
DN 100	8 x M16	70	60
DN 150	8 x M20	140	120

*1 Slightly lubricated threads

*2 Well-lubricated threads

1. Fit an isolating valve to the tank.

The isolating valves can be made of cast iron or PVC. The end of the PVC valve can be pushed directly into the socket with a plug-in seal.

2. Connect the pump to the isolating valve.

3. Fasten the pump to the floor using the dowels and screws supplied with the lifting station.

4. Fit the vent flange to the pump outlet flange.

5. Drill a hole in the tank connection for the vent hose.

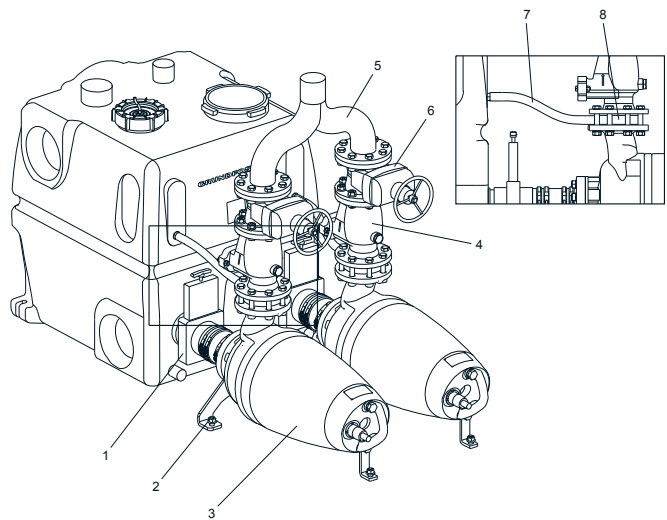
6. Connect the vent hose to the vent flange and to the tank.

7. Fit a non-return valve to the vent flange. Lifting stations must be provided with an approved non-return valve according to EN 12050-4.

8. Fit an isolating valve to the non-return valve.

9. Install the second pump by following the steps above.

Example:



Pos.	Description
1	Isolating valve, accessory
2	Dowels and screws
3	Pump
4	Non-return valve, accessory
5	Breeches pipe, accessory
6	Isolating valve, accessory
7	Vent hose
8	Vent flange

Related information

- [3.2.2.3 Connecting the outlet pipe](#)

3.2.2.2 Connecting the inlet pipe

- All pipe connections must be flexible to reduce resonance.
- Install an isolating valve in the inlet pipe.

Many installations require an inlet below the standard inlet 700 mm above the floor. You can fit a lip seal on site. Use the following accessories:

Product number	Description
91713755	Hole saw, Ø177
91712026	Centring drill
91071939	Lip seal, DN 150, internal Ø160

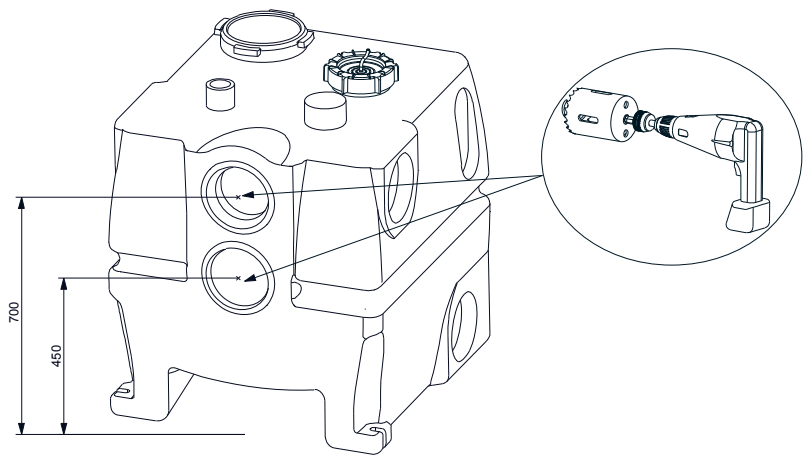


Make sure that the weight from the inlet, outlet and vent pipes does not rest on the tank. Long pipe sections and valves must be supported.



Never stand on the lifting station.

The internal diameter of the inlet pipe must be the same as the internal diameter of the inlet port.
1. Cut open one of the inlet ports to be used and/or connect pipes to the second tank.



2. Fit the DN 150 socket seal in the inlet port or the optional DN 160 socket if the bottom inlet is chosen.
3. Insert the inlet pipe into the socket seal.
 To facilitate maintenance and service of the product, we recommend that you fit an isolating valve between the inlet port and the inlet pipe.
 Required size: DN 100 or DN 150.

Related information

- [2.2 Scope of delivery](#)

3.2.2.3 Connecting the outlet pipe

- All pipe connections must be flexible to reduce resonance.



Make sure that the weight from the inlet, outlet and vent pipes does not rest on the tank. Long pipe sections and valves must be supported.

- To avoid sediment buildup in the outlet pipe, the pipe must be sized for a minimum flow velocity of 0.7 m/s.

Nominal diameter of pipe	80 mm	100 mm
Flow velocity: 0.7 m/s	3.6 l/s	5.6 l/s

- The collecting line must have a filling ratio of at least h/d equal to 0.7.
- The collecting line must be at least one nominal diameter larger after the outlet pipe connection.
- Do not taper off the outlet pipe in the flow direction.
- Lifting stations must be provided with an approved non-return valve according to EN 12050-4.
- The volume of the outlet pipe above the non-return valve up to the backflow level must be smaller than the effective tank volume.
- Install an isolating valve in outlet pipes of DN 80 and up.

1. Connect the outlet pipe to a breeches pipe.
 If you do not install a breeches pipe, make alternative pipework to connect an outlet pipe.
2. Lead the outlet pipe to the collecting line and connect these two.
 The outlet pipe must have a bend above the backflow level which is often the street level.
 Ask local authorities.

3.2.2.4 Connecting the vent pipe

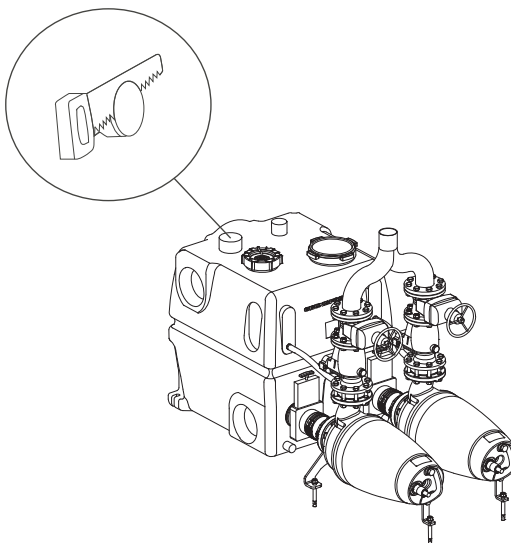
All pipe connections must be flexible to reduce resonance.

Lifting stations for black wastewater must be vented above roof level. However, you may connect the ventilation, as a secondary ventilation, to the main building ventilation system. Place special vent valves outside the building. Special vent valves are available as accessories.



Make sure that the weight from the inlet, outlet and vent pipes does not rest on the tank. Long pipe sections and valves must be supported.

1. Cut off the blanked-off end of the vent port on top of the tank.



2. Fit a DN 70, external Ø75, vent pipe to the vent port by means of the flexible connection piece and two hose clamps.
3. Lead the vent pipe into the main building ventilation system in accordance with local regulations.

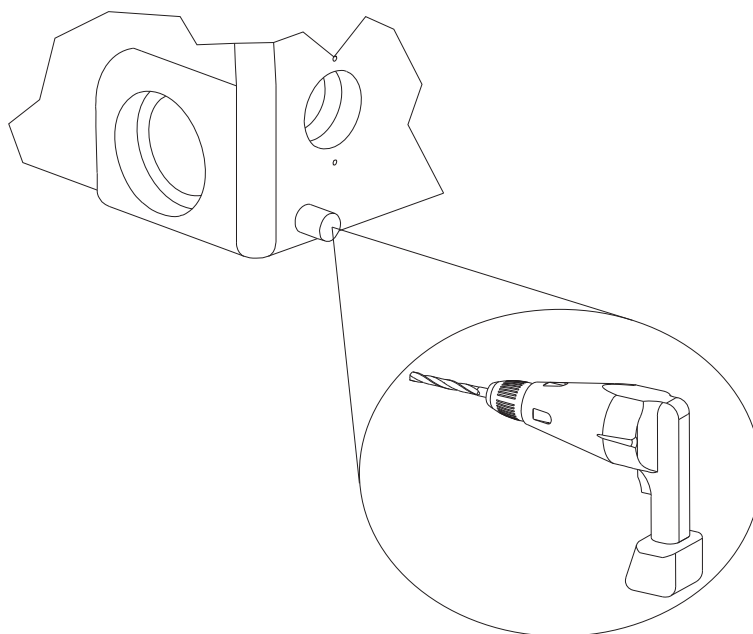
3.2.2.5 Installing a diaphragm pump



Make sure that the weight from the inlet, outlet and vent pipes does not rest on the tank. Long pipe sections and valves must be supported.

You can use the optional, manually operated diaphragm pump for draining the tank before service or maintenance. To enable service or maintenance, we recommend that you fit an isolating valve. The tank has two outlets for 1 1/4" and 1 1/2" pipes. The flexible connecting piece for 1 1/2" is supplied with the lifting station.

1. Drill the outlet port with a 20 mm drill.



It is necessary to drill the outlet as it may have been filled with tank material during the moulding process.

2. Connect the outlet to the pipe for the diaphragm pump by means of the flexible connecting piece and hose clamps.
3. Connect the diaphragm pump to the collecting line.

The outlet pipe of the diaphragm pump must have a bend above the backflow level which is often the street level. Ask your local authorities.

3.2.3 Electrical connection



DANGER **Electric shock**

Death or serious personal injury

- Switch off the power supply before you start any work on the product. Make sure that the power supply cannot be accidentally switched on.

3.2.3.1 Connecting the pumps electrically

Depending on the starting method of the pumps, the control cabinet is supplied with contactors for direct-on-line or star-delta starting.

1. Make sure that the product is suitable for the power supply.
The operating voltage and frequency are marked on the nameplate of the lifting station.
2. Lead all cables through the cable glands of the control cabinet.
3. Connect the pumps according to the wiring diagram supplied with the control cabinet.
4. Set the motor-protective circuit breakers in the control cabinet.

Related information

- [6.1.1 Nameplate](#)

4. Starting up the product

1. Set the selector switches to **OFF**.
2. Switch on the power supply.
It takes up to 45 seconds to boot the controller.
3. When the display is active, select the tank inlet height.
If the inlet height is between two values, select the lower value.
The controller sets the start levels, the stop level and the high-level alarm according to the inlet height selected.
4. Set the rated motor current according to the nameplate.
5. Open the outlet and inlet valves.
6. Set the selector switches to **AUTO**.
7. Flush a toilet or open a tap connected to the tank.
8. Check that the liquid level reaches the start level.
9. Check that the system starts and stops automatically at least twice.

5. Product description

5.1 Applications

The product is designed for the collection and pumping of wastewater with no free flow to sewer level. Typically the product is installed below sewer level in cellars or basements in buildings such as:

- large-scale multifamily houses
- public and commercial buildings
- blocks of flats
- schools
- offices
- hotels
- restaurants
- hospitals
- industries.

5.2 Pumped liquids

The product is designed for the collection and pumping of these liquids:

- wastewater
- sludge-containing water
- domestic wastewater containing faecal matter.

The free passage of the product enables pumping of liquids containing solids from 65 to 80 mm in diameter, depending on the pump type.

Do not discharge these substances or types of wastewater via the product:

- Solid matter, tar, sand, cement, ash, coarse paper, paper towels, cardboard, debris, garbage, grease and oil.
- Wastewater from sanitary installations situated below the backflow level must be drained away via a free-flow drainage system according to EN 12056-1.
- Wastewater containing hazardous substances such as greasy wastewater from large-scale catering establishments.
For drainage of greasy wastewater, use a grease separator according to EN 1825.
- Rainwater of large areas outside the building, for example parking lots, must be collected and discharged outside the building with separate pumping stations.

Related information

- [13.1 Operating conditions](#)

5.3 Lifting station

5.3.1 Tank

The gas-, odour- and pressure-tight tank is made of resistant polyethylene. The tank has all necessary ports for the connection of inlet pipes, outlet pipe, vent pipe and a manually operated diaphragm pump available as an accessory.

Inside the tank the two pump connections are connected to a suction bend enabling suction from the bottom. As a result, the residual volume is reduced to a minimum.

The table below shows the tank volume and the effective volume. The effective volume is the volume between start and stop.

Number of tanks	1	2	3
Tank volume [l]	450	900	1350
Effective volume with stop delay [l]	225	450	675
Effective volume without stop delay [l]	150	300	450

5.3.2 Pumps

The pumps are standard wastewater pumps with supporting feet for horizontal installation. The pumps come in two versions:

Multilift system	Pump version
MD1	SE1, SL1 pump with single-channel impeller
MDV	SEV, SLV pump with SuperVortex impeller

5.3.3 Operating modes

SE pumps

- Continuous operation: The pump runs continuously.
- Intermittent operation: Within a duty cycle of 1 minute, the pump must be stopped for 30 seconds to cool down.

SL pumps

- Intermittent operation: Within a duty cycle of 1 minute, the pump must be stopped for 30 seconds to cool down. The maximum ambient temperature is 30 °C.

We recommend maximum 30 starts per hour for both pump types. However, 60 starts per hour within a two-hour period at peak load are allowed. The purpose is to minimise wear of the shaft seal, bearings and motor.

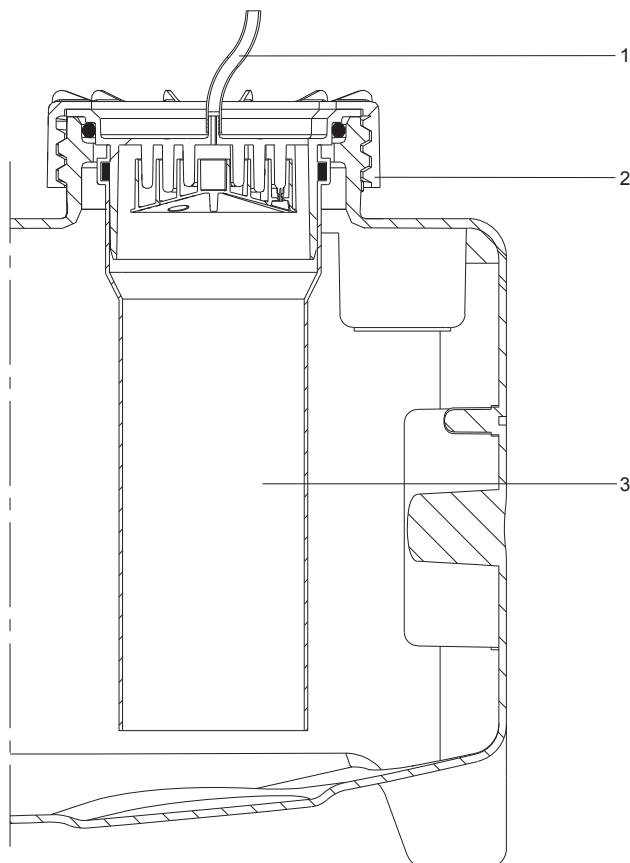
For further information, see the installation and operating instructions for the pumps.

5.3.4 Motor protection

A thermal switch is incorporated in the motor windings. A motor-protective circuit breaker is incorporated in the control cabinet, and the controller has an additional current measurement function.

5.3.5 Level sensor

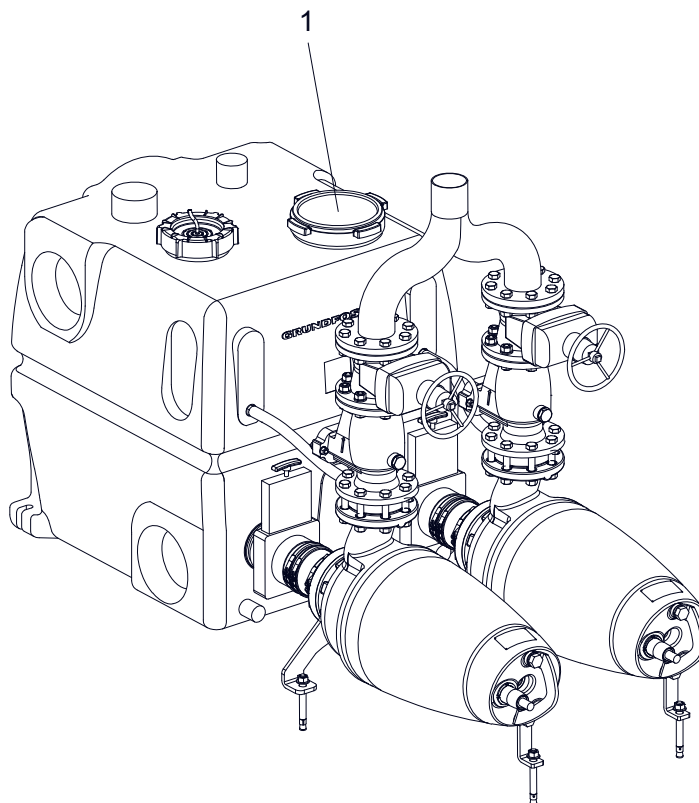
The level sensor consists of a piezoresistive pressure sensor and a pressure tube connected by a pressure hose. The pressure tube is located underneath the screw cap and extends down into the tank. The screw cap includes a condensate trap and a connection for the tube. You can remove the pressure tube for maintenance, service and cleaning. The pressure sensor measures the liquid level in the tank. The rising liquid level compresses the air inside the pressure tube and hose, and the piezoresistive pressure sensor transforms the changing pressure into an analog signal. The controller uses the analog signal to start and stop the pump and to indicate high-level alarm.



Pos.	Description
1	Hose connecting the pressure tube with the pressure sensor
2	Screw cap
3	Pressure tube

5.3.6 Inspection cover

The inspection cover enables inspection of the tank during operation.



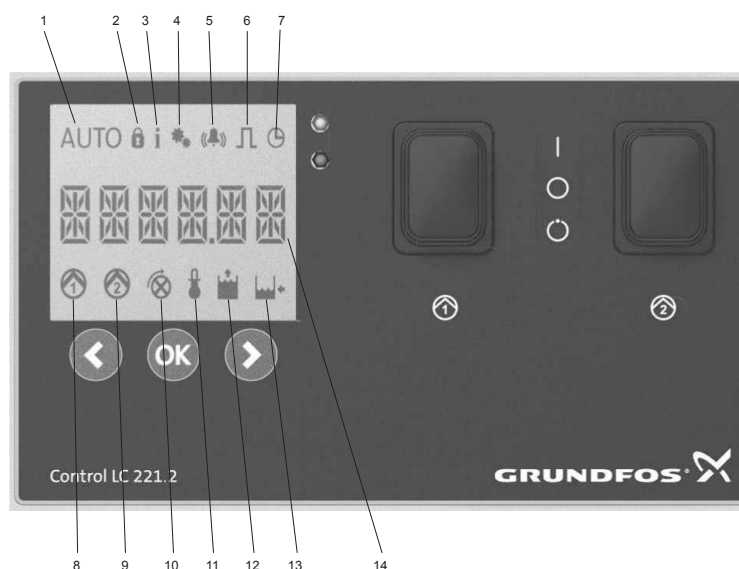
Pos.	Description
1	Inspection cover

5.3.7 Non-return valve

Lifting stations must be provided with an approved non-return valve according to EN 12050-1 and must be approved according to EN 12050-4. The non-return valve must have a correct free passage and a lifting device, for example a screw, to drain the outlet pipe back into the tank.

5.4 Controller

5.4.1 Display



Pos.	Description
1	Automatic mode The symbol is on when the selector switch is in position AUTO .
2	Settings locked You can prevent unauthorised persons from making changes to the settings by locking the controller. To unlock the controller, enter this code: 1234.
3	Information menu Information about fault, operating hours, number of starts and average motor current.
4	Settings menu Information about and setting of start level, rated current, start, stop and alarm delays, maintenance interval and automatic or manual resetting of alarms.
5	Alarm The alarm code appears from the information menu.
6	Impulse counter When the symbol is on, the display shows the number of starts.
7	Settable times and fault indication When the symbol is on, the display shows the number of operating hours, start, stop or alarm delays.
8	Pump 1 The symbol is on when pump 1 is running, and the symbol flashes if pump 1 has a fault.
9	Pump 2 The symbol is on when pump 2 is running, and the symbol flashes if pump 2 has a fault.
10	Phase-sequence fault The symbol applies only to three-phase pumps. The symbol flashes if the phase sequence is wrong or a phase is missing.
11	Overheating The symbol is on if the motor temperature exceeds the permissible value, and the thermal switch has cut out the pump.

Pos.	Description
12	High-level alarm The symbol is on if the liquid level has reached the maximum level.
13	Liquid level When the symbol is on, the display shows the liquid level.
14	Values in the form of digits The value of the function selected.

5.4.2 Buttons



Pos.	Description
1	Press this button to go left in the main menu, up in the vertical menus or decrease a value.
2	Press this button to activate the submenus, confirm the selection or reset the buzzer.
3	Press this button to go right in the main menu, down in the vertical menus or increase a value.

5.4.3 Selector switches



Pos.	Description
1	Selector switch, pump 1
2	Selector switch, pump 2
3	ON Set the selector switch to this position to start the pumps manually.
4	OFF Set the selector switch to this position to stop the pumps and switch off the power supply to the pumps.
5	AUTO Set the selector switch to this position to enable automatic mode.

5.4.4 Status LEDs



Pos.	Description
1	The LED is green when the system is running.
2	The LED flashes red when there is a fault.

5.4.5 Menus

There are two menus in the controller:

- settings menu
- information menu

Settings menu

Menu	Designation	Description
L_01	Start level	Set the start level in this menu.
I_02	Rated current	Set the rated current.
T_01	Stop delay	Set a stop delay for the pumps.
T_02	Start delay	Set a start delay for the pumps.
T_03	Alarm delay	Set an alarm delay for the system.
M_01	Maintenance interval	Set the maintenance interval to 0, 3, 6 or 12 months. To disable the function, set the interval to 0 months.
A_01	Resetting of alarms	Select manual or automatic resetting of alarms.
G_01	Resetting to factory settings	-
Exit	Exit	Go back to the main menu.

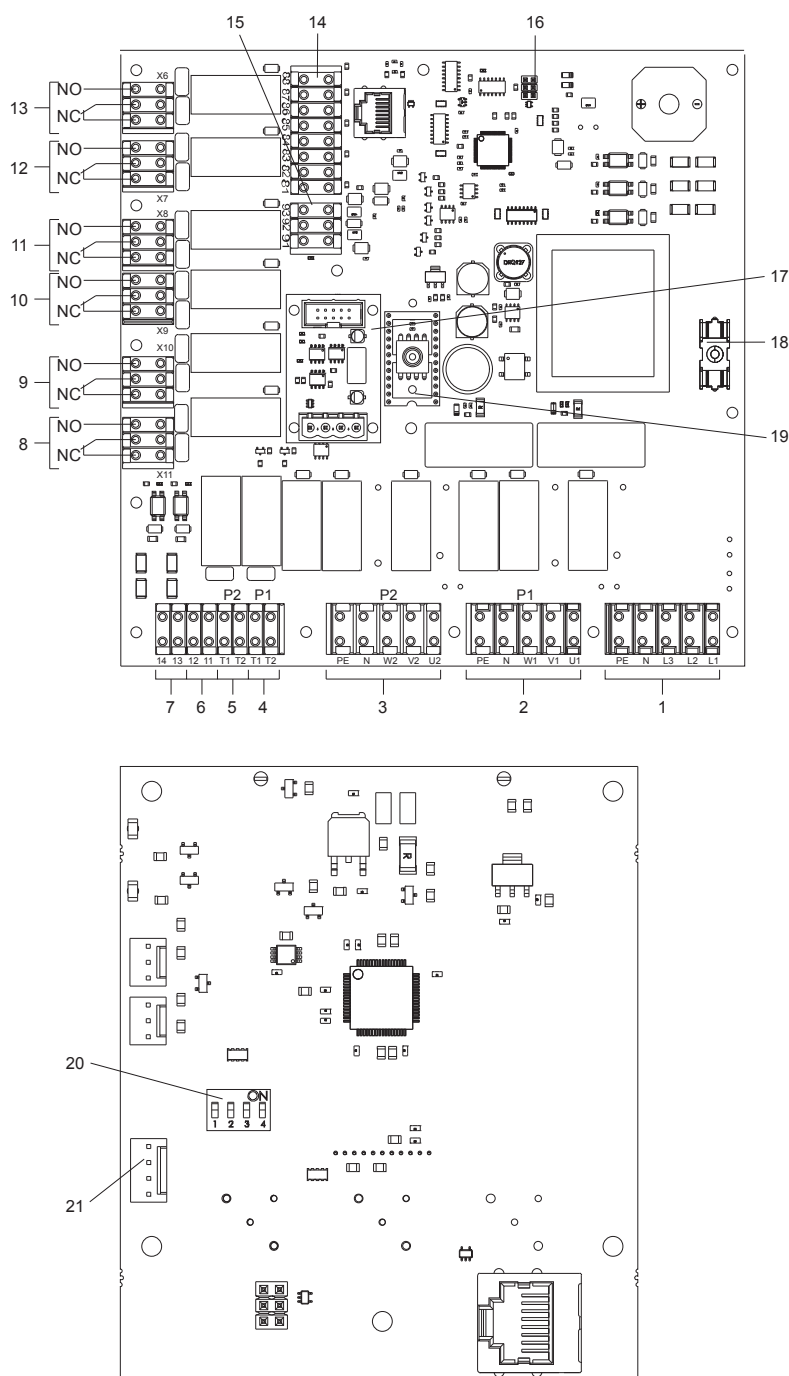
Information menu

Menu	Designation	Description
F_01	Fault log	The fault log shows the latest 20 faults.
R_01	Hours counter [h]	The hours counter shows the number of operating hours.
R_02	Impulse counter	The impulse counter shows the number of starts.
R_03	Motor current [A]	The motor current is the average of the last 20 measurements.
Exit	Exit	Go back to the main menu.

5.4.6 Factory settings

Function	Factory setting
Start level	Not set from factory.
Rated current	Not set from factory; to be set according to the motor nameplate.
Stop delay	1 second
Start delay	0 seconds
Alarm delay	5 seconds
Maintenance interval	0 months

5.4.7 Overview of input and output terminals



Pos.	Terminal	Description
1	PE, N, L3, L2, L1	Connection to power supply
2	PE, N, W1, V1, U1	Connection, pump 1
3	PE, N, W2, V2, U2	Connection, pump 2
4	T1, T2	Thermal switch, pump 1
5	T1, T2	Thermal switch, pump 2
6	11, 12	Reset of external alarm buzzer
7	13, 14	External alarm
8	X11	Common fault
9	X10	High-level alarm
10	X9	Fault, pump 2

Pos.	Terminal	Description
11	X8	Fault, pump 1
12	X7	Operation, pump 2
13	X6	Operation, pump 1
14	81-88	Level switches
14	81, 82	High-level alarm ^{*1}
15	91 (GND), 92 (signal), 93 (12 V)	Analog sensors
16	-	Connection, PC Tool
17	-	Not used
18	0.1 A T	Control-current fuse
19	-	Piezoresistive pressure sensor module
20	-	DIP switches, not used
21	-	Connection, non-rechargeable 9 V battery

^{*1} The high-level alarm can be used as a level detection backup when installed in parallel to an analog level sensor. The high-level alarm starts and stops the pump by float switch hysteresis with a factory-set stop delay of 60 seconds. You can adjust the stop delay by means of Grundfos PC Tool.

5.5 Optional equipment

5.5.1 Accessories

For available accessories, see the link or QR code below.

<http://net.grundfos.com/qr/i/98288126>



5.5.2 Drainage pump

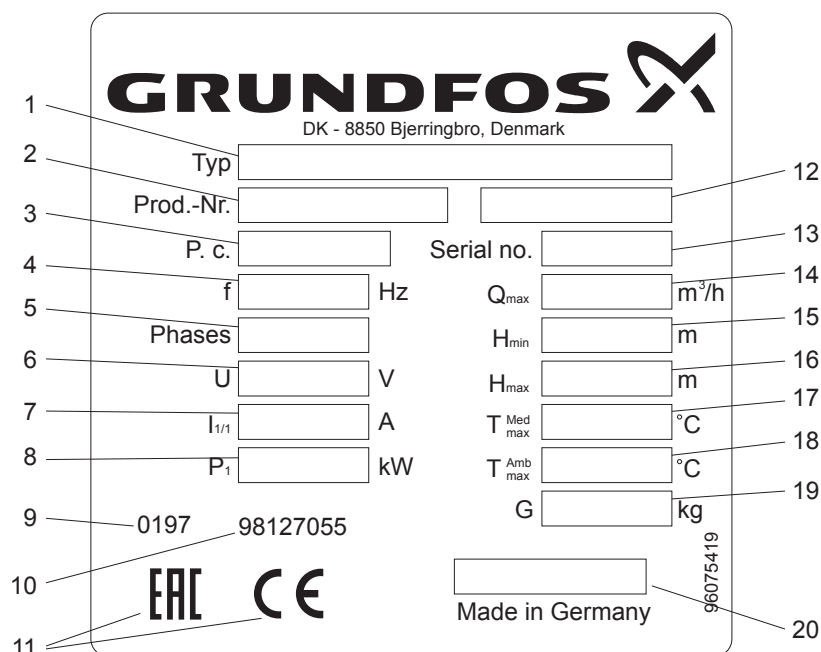
If the lifting station is installed in a basement with penetrating groundwater, we recommend that you install a drainage pump in a separate pump sump below floor level. In some countries, a drainage pump or only a pump sump is required.

6. Identification

6.1 Lifting station

6.1.1 Nameplate

The nameplate is located on the tank.



Pos.	Description
1	Type designation
2	Product number
3	Production code, year and week
4	Frequency
5	Number of phases
6	Rated voltage
7	Full-load current
8	Motor input power, P1
9	Notified body
10	Reference number for the declaration of performance
11	EAC and CE marks
12	Duty type
13	Serial number
14	Maximum flow rate
15	Minimum head
16	Maximum head
17	Maximum liquid temperature
18	Maximum ambient temperature
19	Weight
20	Standard used: Construction Products Directive 89/106/EEC

6.1.2 Type key

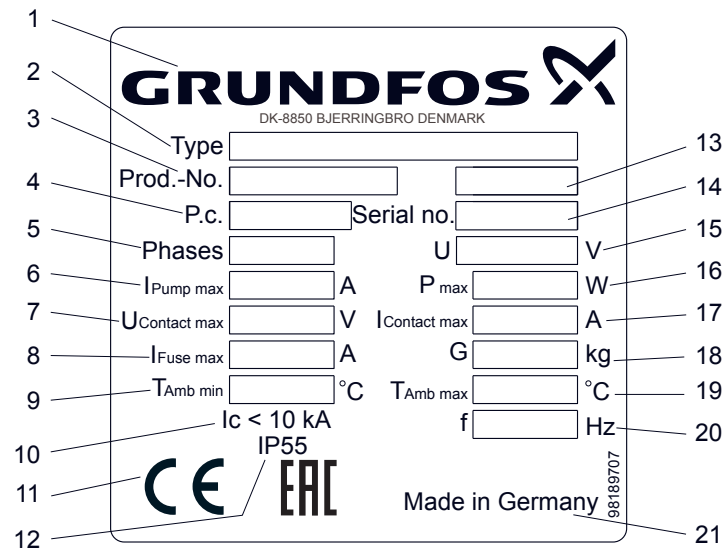
Example: MD1.80.100.15.4.50D/450-2 SE

Code	Designation	Explanation
M	Type range	Multilift lifting station
D	Number of pumps	Two pumps
1	Impeller type	Single-channel impeller
V		SuperVortex impeller
80	Free passage [mm]	Maximum solids size
100	Pump outlet [mm]	Nominal diameter of pump outlet
15	Power [W]	Output power, P2/100
2	Number of poles and speed	2-pole, 3000 min-1
4		4-pole, 1500 min-1
5	Frequency [Hz]	50
6		60
0D	Voltage and starting method	380-415 V, direct-on-line
1D		380-415 V, star-delta
0E		220-240 V, direct-on-line
1E		220-240 V, star-delta
450	Tank volume	450 litres
[]	Number of tanks	One tank
2		Two tanks
SE	Pump type	SE pump
SL		SL pump

6.2 Controller

6.2.1 Nameplate

The nameplate is located on the side of the control cabinet.



Pos.	Description
1	Company address
2	Type designation
3	Product number
4	Production code, year and week
5	Number of phases
6	Maximum pump input current
7	Maximum voltage at potential-free contact
8	Maximum backup fuse
9	Minimum ambient temperature
10	Rated breaking capacity of the motor protection characteristic C
11	CE and EAC marks
12	Enclosure class
13	Version
14	Serial number
15	Rated voltage
16	Maximum power consumption
17	Maximum current at potential-free contact
18	Weight
19	Maximum ambient temperature
20	Frequency
21	Country of origin

6.2.2 Type key

Example: LC 221.2.230.1.10.30 SD

Code	Designation	Explanation
LC 221	Controller type	Level controller
1	Number of pumps	One pump
2		Two pumps
230	Voltage [V]	230
1	Number of phases	Single-phase
3		Three-phase
10	Maximum operating current [A]	-
30/150	Capacitor size, operation and start, [μF]	-
[]	Starting method	Direct-on-line
SD		Star-delta

7. Control functions

7.1 Settings menu

7.1.1 Start and stop levels

The start level of the first pump is preset according to the selected inlet height. The second pump has a higher start level which is activated if the first pump cannot discharge the inflow.

Both pumps stop when the stop level is reached.

The controller switches between the pumps to ensure an even distribution of operating hours.

If the first pump does not start when the first start level is reached, the other pump starts automatically, and the controller display indicates an alarm.

If the first pump stops due to a failure, the second pump starts automatically, and the controller display indicates an alarm.

You can set the start levels in the settings menu.

Related information

- [8.1.1 Setting the start level](#)

7.1.2 Rated current

The purpose of setting the rated current is to give the system the data required to protect the motor.

You must set the rated current during the startup and adjust the motor-protective circuit breaker.

7.1.3 Stop delay

The stop delay is the time from when the stop level is reached until the pump stops. The stop delay reduces water hammer in case of a long inlet pipe.

You can set a stop delay in the settings menu **T_01**.

7.1.4 Pump start delay

The pump start delay is the time from when the start level is reached until the pump starts.

You can set a pump start delay in the settings menu **T_02**.

7.1.5 System start delay

The start delay is the time from when the start level is reached until the system starts.

The factory setting is 45 seconds. You can reduce the delay to 5 seconds by pressing **OK**.

7.1.6 Restarting delay

When restarting the system after an intentional or unintentional power cut, the controller automatically delays the startup by up to 45 seconds. You can reduce the delay to 5 seconds by pressing **OK**. The purpose of the delay is to even out the mains load when several appliances are started up at the same time.

7.1.7 Alarm delay

The alarm delay is the time from when a high-level alarm is detected until an alarm is indicated.

The alarm delay prevents high-level alarm in case of a temporary high inflow to the tank.

You can set an alarm delay in the settings menu **T_03**.

7.1.8 Maintenance interval

You can set the maintenance interval in the settings menu **M_01**.

7.1.9 Factory settings

You can reset the controller to the factory settings in the settings menu **G_01**. This removes any manual settings.

7.2 Information menu

7.2.1 Fault log

The fault log shows the latest 20 faults.

See the fault log in the information menu **F_01**.

7.2.2 Hours counter

The hours counter shows the operating hours for each pump.

See the hours counter in the information menu **R_01**.

7.2.3 Impulse counter

The impulse counter shows the number of starts for each pump.

See the impulse counter in the information menu **R_02**.

7.2.4 Highest measured motor current

The menu shows the highest measured current to each motor.

See the highest measured motor current in the information menu **R_03**.

8. Operating the product

8.1 Setting the controller

8.1.1 Setting the start level

When you switch on the controller for the first time, the startup wizard guides you through the necessary settings. If you later want to change the start level, follow these steps:

1. Go to the settings menu.
2. Press **OK**.
3. Go to menu **L_01**.
4. Press **OK**.
5. Select the start level.
 - Select 450 mm if the low inlet is used.
 - Select 700 mm if the high inlet is used.

Select 450 mm if the inlet pipe height is between the two values.
6. Press **OK**.

8.1.2 Setting the rated current

When you switch on the controller for the first time, the startup wizard guides you through the necessary settings. If you later want to change the rated motor current, follow these steps:

1. Go to the settings menu.
2. Press **OK**.
3. Go to menu **I_02**.
4. Press **OK**.
5. Press arrow right or left to set the rated current according to the motor nameplate.
6. Press **OK**.

8.1.3 Setting the stop delay

1. Go to the settings menu.
2. Press **OK**.
3. Go to menu **T_01**.
4. Press **OK**.
5. Press arrow right or left to set the delay in seconds.
6. Press **OK**.

8.1.4 Setting the start delay

1. Go to the settings menu.
2. Press **OK**.
3. Go to menu **T_02**.
4. Press **OK**.
5. Press arrow right or left to set the delay in seconds.
The factory setting is 0.
6. Press **OK**.

8.1.5 Setting the alarm delay

1. Go to the settings menu.
2. Press **OK**.
3. Go to menu **T_03**.
4. Press **OK**.
5. Press arrow right or left to set the delay in seconds.
6. Press **OK**.

8.1.6 Setting the maintenance interval

1. Go to the settings menu.
2. Press **OK**.
3. Go to menu **M_01**.
4. Press **OK**.
5. Press arrow right or left to select the interval:
 - 0 months
 - 3 months
 - 6 months
 - 12 months.
6. Press **OK**.

8.1.7 Setting of manual or automatic resetting of alarms

1. Go to the settings menu.
2. Press **OK**.
3. Go to menu **A_01**.
4. Press **OK**.
5. Press arrow right or left to select how to reset alarms:
 - MAN
 - AUTO.
6. Press **OK**.

8.1.8 Resetting the controller to the factory settings

1. Go to the settings menu.
 2. Press **OK**.
 3. Go to menu **G_01**.
 4. Press **OK**.
- When you have reset the controller, the startup wizard starts.

9. Servicing the product



DANGER

Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product. Make sure that the power supply cannot be accidentally switched on.



WARNING

Biological hazard

Death or serious personal injury

- Flush the lifting station thoroughly with clean water and drain the outlet pipes. Close the isolating valves. Rinse the dismantled parts in water. Wear personal protective equipment.



CAUTION

Hot surface

Minor or moderate personal injury

- The surface of the product may be so hot that it will cause burns or personal injury.



CAUTION

Back injury

Minor or moderate personal injury

- Use lifting equipment.



CAUTION

Crushing of feet

Minor or moderate personal injury

- Wear safety shoes when moving the product.

9.1 Maintenance schedule

9.1.1 Maintenance intervals

According to EN 12056-4, the lifting station must be checked at the following intervals to ensure safe and reliable operation:

- every 12 months in single-family houses
- every 6 months in multifamily houses
- every 3 months in commercial and industrial applications.

The product may be installed in environments requiring shorter maintenance intervals.

Related information

- [8.1.6 Setting the maintenance interval](#)

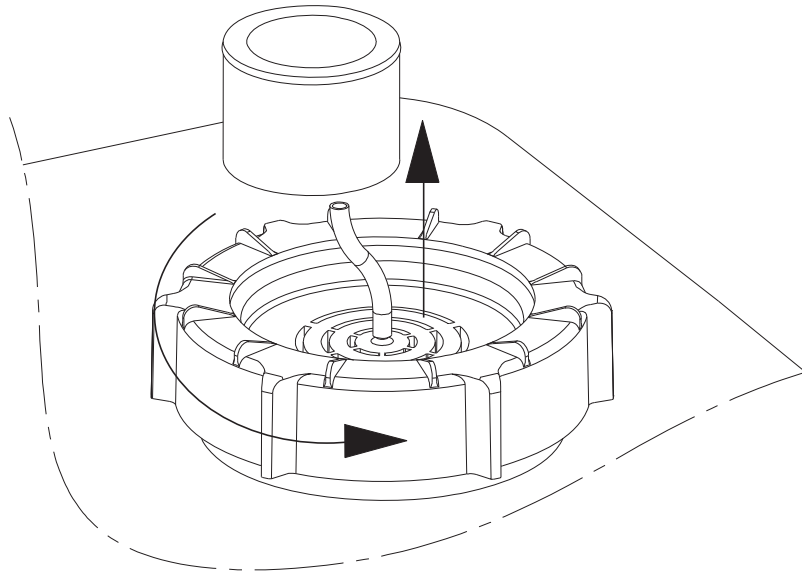
9.2 Mechanical maintenance

9.2.1 Maintaining the product

- Check that all pipe connections are tight.
- Fix any leakages.
- Remove any deposits or sludge buildup in the tank.
- Check the tank for cracks and deformations.
- Remove any blockages on the inlet side of the lifting station.
- Check the pump according to the installation and operating instructions for the pump.

9.2.2 Cleaning the pressure tube and hose

1. Set the selector switches to **OFF**.
2. Close all inlet valves.
3. Loosen the screw cap.



4. Pull the pressure tube carefully out of the tank.
5. Remove any deposits. If necessary, disconnect the pressure hose from the pressure sensor, and rinse the tube and hose with clean water at low pressure. Make sure that no water remains in the pressure hose.
6. Fit the pressure tube in the tank, and tighten the screw cap.
7. Connect the pressure hose to the pressure sensor.
8. Set the selector switches to **AUTO**.

Related information

- [5.3.5 Level sensor](#)

9.3 Electrical maintenance

- Check the gasket of the control cabinet front.
- Check the cable entries in the control cabinet.
- Check the cable connections.
- Check the pressure hose connections.
- Check the controller functions.
- Check the terminals on the main board for possible corrosion.
- If a non-rechargeable 9 V battery is fitted, replace it once a year.

Related information

- [9.2.2 Cleaning the pressure tube and hose](#)

10. Returning the product for service

1. If the product cannot be serviced on site, send it to Grundfos or an authorised service workshop.
Costs of returning the product for service are to be paid by the customer.
2. Before you return the product for service, contact Grundfos or the service workshop with details about the pumped liquid. Otherwise, Grundfos or the service workshop may refuse to receive the product.
3. Thoroughly clean the product before returning it.

11. Taking the product out of operation

**CAUTION****Hot surface**

Minor or moderate personal injury

- The surface of the product may be so hot that it will cause burns or personal injury.

1. Close the inlet valves.
2. Empty the tank:
 - a. Run one pump manually.
 - b. Stop the pump.
 - c. Use a diaphragm pump to empty the tank below the pump inlet height.
3. Close the outlet valves.
4. Switch off the power to the pumps and the controller.

12. Fault finding the product



DANGER
Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product. Make sure that the power supply cannot be accidentally switched on.



WARNING
Biological hazard

Death or serious personal injury

- Flush the lifting station thoroughly with clean water and drain the outlet pipes. Close the isolating valves. Rinse the dismantled parts in water. Wear personal protective equipment.



CAUTION
Hot surface

Minor or moderate personal injury

- The surface of the product may be so hot that it will cause burns or personal injury.



CAUTION
Back injury

Minor or moderate personal injury

- Use lifting equipment.



CAUTION
Crushing of feet

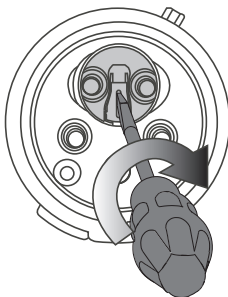
Minor or moderate personal injury

- Wear safety shoes when moving the product.

12.1 F001

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **F001**.
- The symbol of the pump is flashing.
- The symbol of the phase sequence is flashing.

Cause	Remedy
Wrong phase sequence	Change the phase sequence with the phase inverter in the CEE plug.
	Interchange two conductors.



12.2 F002

Conditions:

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **F002**.
- The symbol of the phase sequence is flashing.

Cause	Remedy
Missing phase	Connect the missing phase.

12.3 F003

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **F003**.
- The symbol of the high-level alarm is flashing.
- Both pumps are running.

This alarm indicates high liquid level in the tank. The inflow is higher than the outflow.

Cause	Remedy
Not all outlet valves are open.	Open all outlet valves.
There is a blockage in the tank or pump.	Remove the blockage.
The pump is not vented correctly. The pump cannot build up pressure.	1. Check that the vent flange is correctly fitted between the pump outlet and the outlet valve. 2. Check that the vent hose is correctly fitted between the vent flange and the tank. 3. Remove any blockage from the vent pipe or vent flange.
The lifting station is under-sized.	Recalculate the inflow parameters and compare the result with the tank volume and pump performance. If you need a new product, contact the nearest Grundfos sales company.

Related information

- [3.2.2.1 Installing the pumps](#)

12.4 F007, F008

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- The symbol of the pump is flashing.
- The faulty pump stops, and the other pump starts.
- Display text: **F007** or **F008**.

The controller has detected overcurrent and stopped the faulty pump.

Fault code	Cause	Remedy
F007	Pump 1 is blocked.	Remove the blockage.
	Missing phase, pump 1.	1. Check the motor windings. 2. Connect the missing phase.
F008	Pump 2 is blocked.	Remove the blockage.
	Missing phase, pump 2.	1. Check the motor windings. 2. Connect the missing phase.

12.5 F117

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **F117**.

Cause	Remedy
The cable between the main board and the display is not connected properly.	Disconnect and reconnect the cable.
The cable between the main board and the display is not connected properly or is defective.	Replace the cable. Use an ethernet cable, RJ45.
The problem is still not solved. The connectors between the main board and the display may be defective.	Contact the nearest Grundfos sales company.

12.6 RELAY

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **RELAY**.
- The pump does not start or stop.
- The symbol of the pump is flashing.

Find the cause of the alarm by means of the fault code in the information menu **F_01**.

Fault code	Cause	Remedy
F015	The relay or contactor for pump 1 is defective. The pump does not stop.	Solution 1
F017	The relay or contactor for pump 2 is defective. The pump does not stop.	
F016	The relay or contactor for pump 1 does not close. The pump does not start. The motor-protective circuit breaker has tripped, or the main board is defective.	Solution 2
F018	The relay or contactor for pump 2 does not close. The pump does not start. The motor-protective circuit breaker has tripped, or the main board is defective.	

Solution 1

1. Switch off the power supply to stop the pump.
2. Replace the defective relay or contactor.

Solution 2

1. Reset the motor-protective circuit breaker.
2. Start the pump manually and make sure that it is running.
3. If resetting the motor-protective circuit breaker does not solve the problem, contact the nearest Grundfos sales company.

12.7 TEMP

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- The temperature symbol is flashing.
- The symbol of the pump is flashing.
- The faulty pump stops, and the other pump starts.
- Display text: **TEMP**.

The pump has started too many times and is overheated. The thermal switch has stopped the pump.

Cause	Remedy
Leakage or incorrect sizing of the lifting station.	1. Check the non-return valve for any leakage. 2. If there is no leakage, check the inflow parameters. The inflow must not be higher than the pump performance. If the inflow is too high, contact the nearest Grundfos sales company or reduce the inflow. 3. Allow the pump to cool. The pump restarts automatically.

12.8 SENSOR

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **SENSOR**.

The alarm indicates that the sensor signal is out of range. Go to the information menu **F_01**.

1. If the fault code **F003** appears on top of the list, there is a high-level alarm. The inflow is higher than the outflow, and the liquid level is above the sensor measuring range of 1000 mm. The pumps have been started. Go to solution 1.
2. If **F003** is not one of the latest faults, the sensor type could have accidentally been changed via the display or Grundfos PC Tool. Go to solution 2.
3. If the problem is still not solved, contact the nearest Grundfos sales company.

Solution 1

Cause	Remedy
Not all outlet valves are open.	Open all outlet valves.
There is a blockage in the tank or pump.	Remove the blockage.
The pump is not vented correctly. The pump cannot build up pressure.	1. Check that the vent flange is correctly fitted between the pump outlet and the outlet valve. 2. Check that the vent hose is correctly fitted between the vent flange and the tank. 3. Remove any blockage from the vent pipe or vent flange.
The lifting station is under-sized.	Recalculate the inflow parameters and compare the result with the tank volume and pump performance. If you need a new product, contact the nearest Grundfos sales company.

Solution 2

Cause	Remedy
The sensor type has accidentally been changed.	Reset the controller to the factory settings in the settings menu G_01 .

Related information

- [3.2.2.1 Installing the pumps](#)

12.9 EXTERN

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **EXTERN**.

Cause	Remedy
Flooding outside the tank. The alarm is related to the external level switch connected to the controller.	Find the source of the flooding and eliminate the problem.

12.10 BATT

- The red LED is flashing.
- The buzzer and the alarm symbol are on.
- Display text: **BATT**.

Cause	Remedy
The battery is discharged.	Replace the battery. Use a non-rechargeable battery.

12.11 The pumps do not start

- The power supply is switched on.
- The selector switches are in position **AUTO** or **ON**.

The display is off

Cause	Remedy
The control-current fuse is blown.	Solution 1
The supply cable to the controller is defective.	Solution 2
The connections of the controller supply cable are loose.	Solution 3
The main board is defective.	Solution 6
The cable between the main board and the display is not connected properly or is defective.	Solution 7
The problem is still not solved.	Solution 4

The display is on

Cause	Remedy
The pump supply cable is not connected properly to the controller.	Solution 3
The pump supply cable is defective.	Solution 2

The display indicates a liquid level which is lower than the level in the tank

Cause	Remedy
There is a leakage in the level sensor.	Solution 5

Solution 1

1. Find and eliminate the cause.
2. Replace the fuse.

Solution 2

Replace the cable.

Contact the nearest Grundfos sales company.

Solution 3

Tighten the cable connections.

Solution 4

The controller is defective.

Contact the nearest Grundfos sales company.

Solution 5

- Check the connection between the pressure tube and the pressure hose.
- Check the connection between the pressure hose and the pneumatic coupling underneath the controller.
The hose is connected correctly when you cannot pull it out without pressing the locking mechanism.
- Check the connection between the pressure hose and the pressure sensor on the main board.

Solution 6

1. Check that all the connections to the main board are tight.
2. If not, the main board is defective and you must replace it.
Contact the nearest Grundfos sales company.

Solution 7

- 1. Check that the cable is connected properly at both ends.
- 2. If the problem is still not solved, replace the cable.
Use an ethernet cable, RJ45.

Related information

- [9.2.2 Cleaning the pressure tube and hose](#)

12.12 The pumps start randomly

- No alarm

Cause	Remedy
Automatic test run 24 hours after last operation.	No action necessary. It is a safety function that prevents the shaft seal from seizing up.

Related information

- [9.2.2 Cleaning the pressure tube and hose](#)

13. Technical data

13.1 Operating conditions

Controller

Ambient temperature during operation [°C]	0-40
---	------

Pump

Ambient temperature [°C]	SE: -20 to +40 SL: -20 to +30
Liquid temperature [°C]	0-40 For short periods up to 60
pH value	4-10
Maximum density [kg/m ³]	1100
Sound pressure level	Less than 70 dB (A)

13.2 Mechanical data

Control cabinet

Ambient temperature in stock [°C]	-30 to +60
Height x width x depth [mm]	Direct-on-line starting: 390 x 262 x 142 Star-delta starting: 680 x 380 x 350
Weight	See the controller nameplate.
Enclosure class	Direct-on-line starting: IP55 Star-delta starting: IP54

Related information

- [6.2.1 Nameplate](#)

13.3 Electrical data

Power supply

Supply voltage [V]	3 x 380-415, PE 3 x 220-240, PE
Voltage tolerances	- 10 %/+ 6 % of rated voltage
Frequency [Hz]	50 or 60
Supply system earthing	TN system
Backup fuse	See the controller nameplate.

Controller

Voltage tolerances	- 10 %/+ 6 % of rated voltage
Frequency [Hz]	50 or 60
Control-current fuse	Fine-wire fuse, 100 mA, 250 V, 20 mm x Ø5
Outputs for alarm devices	Potential-free contacts, NO/NC, maximum 250 VAC, 2 A
Input, external reset	230 V
Power consumption [W]	Approximately 7

Related information

- [6.2.1 Nameplate](#)

14. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way.

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

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